Maxwell Fang

20941185 • + 1 (647) 636-8837 • maxwellfang63@gmail.com

SKILLS

Language Skills: Advanced in English, intermediate in French, basic in Mandarin

Programming Languages: **Python**, **SQL**, **R**, **Matlab**, **Bash**

Tools: **Git, Docker, Kubernetes, AWS EC2, EKS, ECR, S3, Gstreamer, Microsoft Office, Tableau**

RDBMS: MS SQL Server Management studio, SQL Developer, Microsoft SQL server 19

PROFESSIONAL & LEADERSHIP EXPERIENCE

University of Waterloo Waterloo, Ontario

Undergraduate Research Assistant (Volunteer) under S. Vavasis Nov 2024 – Present

- Generated artificial image datasets using **numpy** and **OpenCV** that modeled neural scaling laws to analyze eigenpairs of the covariance matrix
- Conducted numerical experiments on MNIST and Artificial datasets to understand and determine causes of eigenvector localization in Fourier transformed covariance matrices in artificial datasets

Ford Motor Company of Canada Waterloo, Ontario

Software Developer – AI/ML Jan 2025 – April 2025

- Designed and implemented Perspective View to Bird's Eye View image transformation module and inference visualization using LIDAR and RADAR ground truths with **Python**, **cv2**, and **numpy**
- Developed image interpolation to correct visual artifacts in post-transformation Bird's Eye View images and motion smoothing to augment low framerate videos
- Created visualization pipeline using **PyGObject Gstreamer** and **threading** to evaluate accuracy of LIDAR/RADAR ground truths and lane-line prediction model by visualizing inferences on PV images and converting to motion-smoothed video

Eon Media Toronto, Ontario

Lead AI/ML Engineer Co-op May 2024 – August 2024

- Led and trained a team of 3 coops to maintain pre-existing video processing pipelines, object detection models, NLP models, and develop AI/ML solutions
- Implemented **Robust Continuous Clustering Dimensionality Reduction** with **Conjugate Gradient Method** and **PCA** to cluster high dimensional image embeddings (2048 dimensions) created by **Pytorch resnet50** and reduce former clustering runtime 75% using **numpy**, **scipy.sparse** and **concurrent.futures**
- Designed and implemented Logo detection model training pipeline to reduce manual labour by 45% and 3rd-party software costs by 80%, featuring image clustering, data cleaning, formatting, and augmentation for **YOLO**-based models

Eon Media Toronto, Ontario

AI/ML Engineer Co-op January 2024 – April 2024

- Optimized and redesigned video processing pipelines using Python **multiprocessing** and **Kubernetes** to decrease CPU usage by 85%, pipeline setup time by 60% and overall processing time by 40%
- Designed and minimized runtime of pre-existing object detection models to implement a Reverse Image Search that identifies brands, people and objects in images and returns keywords in < 400 ms hosted on **Flask** with **Nginx**, and **Gunicorn**
- Created and Containerized **python** and **CUDA** based applications for metadata extraction from video (brands, people, location, context, etc.) using **Docker**

CAA South Central Ontario Thornhill, Ontario

QA Automation Intern May 2022 - August 2022

ACHIEVEMENTS

Distinctions:

President's Scholarship of Distinction (Application Average +95%)

International Baccalaureate Diploma (IB score 38 out of 45)

Extracurricular:

Euclid Mathematics Contest Mississauga, Ontario | April 2021

Canadian Senior Mathematics Contest Mississauga, Ontario | Nov 2020

EDUCATION

University of Waterloo – Faculty of Mathematics Ontario, Canada

Candidate for Bachelor of Mathematics (Co-op) – 4th Year Sept 2021 - April 2026

Major in Combinatorics and Optimization